COURSE SYLLABUS

BOROUGH OF MANHATTAN COMMUNITY COLLEGE
The City University of New York
Department of Computer Information Systems

Title of Course: **Advanced Programming Techniques Honors** **Class Hours:** 2

CSC 211H **Laboratory Hours per Week:** 3

Semester Spring 2026
Instructor Information (Phone#, Office#, email):

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Credits: 3

Course Description:
This is a second course in programming which will further develop those skills gained in CSC 111 emphasizing reliability, maintainability, and reusability. Students will be introduced to applications of Pointers, Dynamic memory allocation, Arrays, Abstract data types, Objects, classes, and object-oriented design. Additional programming topics such as Inheritance, Polymorphism, Text Processing, Exception Handling, Recursion and Templates will also be covered.

Basic skills: ENG 095; ESL 095; ACR 095

Prerequisites: CSC 111 (Computer Programming I) or departmental approval

Course Student Learning Outcomes (Students will be able to…)	**Measurements (means of assessment for student learning outcomes listed in first column)**
1. Design and develop programs using arrays, pointers and classes.	**1.** Programming projects, lab exercises and exam questions.
2. Design and implement programs effectively considering compile-time and run-time memory allocation.	**2.** Programming projects and lab exercises.
3. Describe inheritance, write inherited classes and give reasons for using inheritance and polymorphism.	**3.** Lab exercises, exam questions and programming projects.
4. Write program that detects and handles exceptions.	**4.** Programming projects and exam questions.
5. Apply object-oriented programming (OOP) techniques to develop applications.	**5.** Programming projects.

Below are the college's general education learning outcomes, the outcomes that are checked in the left-hand column indicate goals that will be covered and assessed in this course. (Check at least one.)

	General Education Learning Outcomes	**Measurements (means of assessment for general education goals listed in first column)**
☐	**Communication Skills-** Students will be able to write, read, listen and speak critically and effectively.	Essay questions and term project/paper.
☐x	**Quantitative Reasoning-** Students will be able to use quantitative skills and the concepts and methods of mathematics to solve problems.	Use formulas and concepts of mathematics to solve problems in programming assignments.

☐	**Scientific Reasoning-** Students will be able to apply the concepts and methods of the natural sciences.	
☐	**Social and Behavioral Sciences-** Students will be able to apply the concepts and methods of the social sciences.	
☐	**Arts & Humanities-** Students will be able to develop knowledge and understanding of the arts and literature through critiques of works of art, music, theatre or literature.	
x	**Information & Technology Literacy-** Students will be able to collect, evaluate and interpret information and effectively use information technologies.	Use a high-level computer programming language to create application software.
☐	**Values-** Students will be able to make informed choices based on an understanding of personal values, human diversity, multicultural awareness and social responsibility.	

Required Text & Readings

Textbook: **Starting out with C++ from Control Structures through Objects, 9/E , 2018**
Author: **Tony Gaddis**
Publisher: **Pearson**
ISBN: **978-0-13-4498379**

Other Resources: Flash drives are required.

Evaluation & Requirements of Students:

Midterm **20%**
Quizzes **15 %**
Final **20%**
HW Assignments **10%**
Honors Project **20%**
Class Participation/(Classroom Lab, Participation Activity, Discussion) /
Instructor Evaluation **15%**
100%

Outline of Topics:
Students should read the text section of the following chapters as well as the exercises. Reading these sections will help you prepare for your homework and examinations. Your instructor will be assigning specific exercises and programming problems from each chapter as homework.

Week No		Topic
Week 1	**Chapter 13**	**Introduction to Classes (Constructor, destructor, member functions)**
Week 2	**Chapter 13**	**Introduction to Classes (Arrays of objects, OOP analysis and design)**
Week 3	**Chapter 7**	**Arrays and Vectors (Multi-dimensional arrays, STL vector)**
Week 4	**Chapter 9**	**Pointers (Dynamic memory allocation & pointers to structures and class objects)**
Week 5	**Chapter 9**	**Pointers (Dynamic memory allocation & pointers to structures and class objects)**
Week 6	**Chapter 12**	**Advanced File operations**
Week 7		**Review & MIDTERM EXAM**

Week 8	**Chapter 14**	**More about classes (Aggregation, composition, and operator overloading)**
Week 9	**Chapter 14**	**More about classes (Aggregation, composition, and operator overloading)**
Week 10	**Chapter 15**	**Inheritance, Polymorphism, and Virtual Functions**
Week 11	**Chapter 15**	**Inheritance, Polymorphism, and Virtual Functions**
Week 12	**Chapter 16**	**Exceptions and Templates**
Week 13	**Chapter 17**	**The Standard Template Library (STL)**
Week 13-14	**Chapter 20**	**Recursion**
Week 14		**Final Exam Review**
Week 15		**Final Exam**

BMCC Computational Thinking Across the Curriculum Project:

This class is participating in BMCC Computational Thinking Across the Curriculum Project funded by the National Science Foundation (NSF) to promote STEM-thinking, interests, and skills in community college students. As part of this class, you will be engaging in activities that develop computational thinking skills which will help you build critical thinking, analytic and problem-solving skills to enhance your learning in this class. We will learn about computational thinking (CT) and engage in CT-based learning activities.
The CT skills and perspective you will learn in the class are valuable for a range of different fields and careers. Your participation in the research component of the study will provide BMCC faculty researchers with valuable data that will be used to create better STEM-learning experiences and opportunities, especially for community college students from underrepresented groups.

Class Participation

Participation in the academic activity of each course is a significant component of the learning process and plays a major role in determining overall student academic achievement. Academic activities may include, but are not limited to, attending class, submitting assignments, engaging in in-class or online activities, taking exams, and/or participating in group work. Each instructor has the right to establish their own class participation policy, and it is each student's responsibility to be familiar with and follow the participation policies for each course.

BMCC is committed to the health and well‑being of all students. It is common for everyone to seek assistance at some point in their life, and there are free and confidential services on campus that can help.

Single Stop www.bmcc.cuny.edu/singlestop, room S230, 212‑220‑8195, singlestop@bmcc.cuny.edu. If you are having problems with food or housing insecurity, finances, health insurance or anything else that might get in the way of your studies at BMCC, come by the Single Stop Office for advice and assistance. Assistance is also available through the Office of Student Affairs, S350, 212‑220‑8130, studentaffairs@bmcc.cuny.edu.

Counseling Center www.bmcc.cuny.edu/counseling, room S343, 212‑220‑8140, counselingcenter@bmcc.cuny.edu. Counselors assist students in addressing psychological and adjustment issues (i.e., depression, anxiety, and relationships) and can help with stress, time management and more. Counselors are available for walk‑in visits.

Office of Compliance and Diversity https://www.bmcc.cuny.edu/about-bmcc/compliance-diversity, room S701, 212-220-1236. BMCC is committed to promoting a diverse and inclusive learning environment free of unlawful discrimination/harassment, including sexual harassment, where all students are treated fairly. For information about BMCC's policies and resources, or to request additional assistance in this area, please visit or call the office, or email olevy@bmcc.cuny.edu, or twade@bmcc.cuny.edu. If you need immediate assistance, please contact BMCC Public safety at 212-220-8080.

Office of Accessibility www.bmcc.cuny.edu/accessibility, room N360 (accessible entrance: 77 Harrison Street), 212-220-8180, accessibility@bmcc.cuny.edu. This office collaborates with students who have documented disabilities, to coordinate support services, reasonable accommodations, and programs that enable equal access to education and college life. To request an accommodation due to a documented disability, please visit, call the office or email.

BMCC Policy on Plagiarism and Academic Integrity Statement

Plagiarism is the presentation of someone else's ideas, words or artistic, scientific, or technical work as one's own creation. Using the idea or work of another is permissible only when the original author is identified. Paraphrasing and summarizing, as well as direct quotations, require citations to the original source. Plagiarism may be intentional or unintentional. Lack of dishonest intent does not necessarily absolve a student of responsibility for plagiarism. Students who are unsure how and when to provide documentation are advised to consult with their instructors. The library has guides designed to help students to appropriately identify a cited work. The full policy can be found on BMCC's Web site, www.bmcc.cuny.edu. For further information on integrity and behavior, please consult the college bulletin (also available online).